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BrainNetTest: Global and Edge-Wise Inference for Binary Brain Network Populations in R

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Abstract

Clinical connectomics often compares one binary brain network per participant and asks both whether diagnostic groups differ globally and which connections carry the difference. We present **BrainNetTest** 1.0.0, an R package for populations of aligned, binary, undirected brain networks. A validated S3 input class enforces the brain-network analysis contract, and a classed result supplies standard `print()`, `summary()`, `plot()`, and `as.data.frame()` methods. The global procedure is a finite-sample label-randomization test based on the edge-separable L_1 analysis-of-variance score of Fraiman and Fraiman. Separate edge-wise Fisher tests use Holm family-wise error control by default. The implementation includes chunked matrix computation and prefix-sum updates for an optional descriptive ablation path. In prespecified simulations, the largest observed complete-null global rejection rate was 0.051 and the largest Holm family-wise error rate was 0.015. Across four alternatives, mean Holm true-positive rate and precision were 0.191 and 0.716. In a real ABIDE resting-state fMRI application, the observed global statistic is extreme under autism-control label randomization in the full cohort and in a covariate-balanced sensitivity sample, while no single edge survives strict multiplicity control. The package source, tests, clinical application, and standalone replication scripts are distributed together.

Keywords: R, brain connectomics, resting-state fMRI, randomization test, edge-wise inference, clinical group comparison.

1. Introduction

Functional and structural connectomics represent each participant by a brain network whose nodes are atlas regions and whose edges encode estimated connections. Clinical studies then compare populations of these networks, for example participants with a diagnosis and matched controls. Two questions are central. First, is diagnosis associated with a global change in the spatial pattern of connections? Second, which individual connections differ after accounting for the large number of tested edges? The first question can reveal a distributed population effect even when the second has limited power; the second is needed before any particular connection is described as selected.

The distinction matters in both software and interpretation in R (R Core Team 2026). The network-based statistic identifies connected components of edge-wise effects under a component-forming threshold (Zalesky, Fornito, and Bullmore 2010); the **NBR** package extends that framework to linear and mixed-effects designs (Gracia-Tabuenca and Alcauter 2020). The **FIAR** package models directed effective connectivity from fMRI time series (Roelstraete and Rosseel 2011), and **brainGraph** supports construction, graph-theoretic analysis, and visualization of brain MRI networks (Watson 2026). The **NetworkComparisonTest** package instead compares networks estimated from tabular observations, including global strength and edge invariance (van Borkulo, van Bork, Boschloo, Kossakowski, Tio, Schoevers, Borsboom, and Waldorp 2023). General graph infrastructure is provided by **igraph** (Csardi and Nepusz 2006). These tools address different stages or hypotheses from the present setting: independent clinical groups whose observations are themselves aligned binary brain networks.

Fraiman and Fraiman proposed an analysis-of-variance score for samples of networks and studied both detection and subnetwork identification (Fraiman and Fraiman 2018). **BrainNetTest** implements a finite-sample randomization version of the global score and provides separate multiplicity-controlled edge inference. It also retains the edge-separable computational idea behind the original package, while restricting its statistical interpretation. Version 1.0.0 is a redesign around four goals:

1. encode and validate the complete brain-network data contract before computation;
2. expose results through familiar R classes and methods;
3. separate global, edge-level, and descriptive-ablation results without conflating their inferential meanings; and
4. make the clinical application, simulations, and computational benchmarks reproducible from standalone scripts.

The paper follows a domain–software–method–application structure. Section 2 states the clinical analysis contract. Section 3 presents the S3 classes and interface. Section 4 defines the statistical and computational contributions. Section 5 gives executable workflows, and Section 6 applies them to a real autism-control resting-state fMRI study. Sections 7 and 8 report simulation validation and measured computational performance. Limitations are discussed in Section 9.

2. Brain-network analysis contract and clinical context

2.1. From images to network populations

An observation is one participant's binary, undirected brain network without self-loops, represented by a symmetric $p \times p$ adjacency matrix with zero diagonal. All participants must use the same labeled and ordered brain atlas. The input contains $m \geq 2$ independent clinical, behavioral, or demographic groups and at least two observations per group. Sample sizes may be unequal.

These matrices are downstream neuroimaging objects. Atlas selection, registration, connectivity estimation, nuisance adjustment, thresholding, and quality control occur before **BrainNetTest**. The package does not treat an estimated connection as free from upstream uncertainty. Users must also ensure that one participant contributes only one independent observation. Family structure, repeated scans, paired designs, or site clusters require restricted permutations or a model that is not implemented in version 1.0.0.

Weighted or directed connections, missing regions, covariates, and paired or clustered samples are rejected rather than silently coerced. The same mathematical contract can be used outside neuroimaging whenever each independent unit yields an aligned binary network, but the software, documentation, and case study are designed around brain-network populations.

2.2. Two inferential questions

The global randomization null is exchangeability of whole participant networks with respect to group labels. In the standard independent-groups setting, this corresponds to equality of the group graph laws. However, the chosen score depends only on marginal edge frequencies. Therefore, the test is targeted to changes in the spatial pattern of edge occurrence and is not an omnibus detector of alternatives that alter only cross-edge dependence, motifs, degree dependence, or other topology while preserving every edge probability.

Edge tests answer a separate clinical question: for each possible anatomical connection, are its presence probabilities equal among groups? Their decisions are not gated by the global result and are not an attribution or decomposition of global rejection. Selected edges are statistical group associations, not causal connections or validated diagnostic biomarkers.

3. Software design and R interface

3.1. S3 data and result classes

The `brainnet_data()` constructor checks the outer group structure, sample sizes, matrix types and dimensions, finite binary entries, symmetry, zero diagonals, and node ordering. It stores optional node labels and community metadata once. The resulting `brainnet_data` object has `print()` and `summary()` methods that avoid printing individual matrices.

The `brainnet_test()` function returns a `brainnet_result`. Its global component is compatible with `htest`; the full object also stores the complete edge family, central graphs, multiplicity settings, randomization metadata, the call, elapsed time, and optional ablation diagnostics. Standard methods and focused extractors provide the user interface:

- `print()` and `summary()` report the global result, multiplicity procedure, and selected-edge count;
- `as.data.frame()` returns every tested edge with group proportions, effect size, raw and adjusted p-values, and decision;

Software	Input unit	Primary target	Distinguishing scope
FIAR	fMRI time series	effective connectivity	directed models
brainGraph	connectivity matrices	graph analysis	MRI workflows
NBR	edge outcomes	connected components	linear/mixed models
NetworkComparisonTest	tabular samples	estimated-network invariance	independent/paired
BrainNetTest	subject brain networks	global plus edge marginals	binary multi-group API

Table 1: Related R software for connectivity and network analysis. The packages operate at different stages of a neuroimaging workflow and target different hypotheses.

- `selected_edges()` and `selected_nodes()` provide focused views, with the node summary explicitly descriptive; and
- `plot()` visualizes group central graphs and multiplicity-controlled selected edges using **igraph**.

The version 1.0.0 source archive is distributed with the submission under the MIT license; its SHA-256 digest is recorded in `package-source.txt`. The archive is the software object reviewed by the replication scripts.

3.2. Comparison with related software

Table 1 places **BrainNetTest** within the main R tools used for brain connectivity and network comparison.

4. Statistical and computational engine

4.1. Central graphs and the global score

Let $\mathcal{E} = \{(i,j) : 1 \le i < j \le p\}$ be the possible undirected edges. For edge e, write $p_{k,e}$ for its sample proportion in group k and $p_e = \sum_k n_k p_{k,e}/n$ for the pooled proportion. Define

$$d_{k,e} = 2p_{k,e}(1 - p_{k,e}), \qquad D_{k,e} = p_{k,e} + p_e - 2p_{k,e}p_e.$$

The package uses the unnormalized Fraiman score

$$T = \sqrt{m}\sum_{k=1}^{m}\sqrt{n_k}\left\{\frac{n_k}{n_k - 1}\sum_{e\in\mathcal{E}} d_{k,e} - \frac{n}{n-1}\sum_{e\in\mathcal{E}} D_{k,e}\right\}. \tag{1}$$

Every undirected edge is counted once. The positive asymptotic normalizer in the original formulation is unnecessary for randomization ordering and is not exposed as a user parameter.

Because the sign under finite unbalanced and multi-group alternatives need not be fixed, version 1.0.0 predeclares the two-sided extremeness statistic $S = |T|$. Group labels are reassigned uniformly while preserving all sample sizes. If all N distinct assignments are enumerated, including the observed assignment,

$$p_{\text{exact}} = \frac{1}{N}\sum_{\pi=1}^{N}\mathbb{I}\{|T_\pi| \ge |T_{\text{obs}}|\}.$$

Automatic exact enumeration is used when $N \leq 10{,}000$. Otherwise, for B random assignments sampled with replacement from the complete fixed-size orbit (including the possibility of drawing the observed assignment),

$$p_{\mathrm{MC}} = \frac{1 + \sum_{b=1}^{B} \mathbb{I}\{|T_b| \geq |T_{\mathrm{obs}}|\}}{B+1}. \tag{2}$$

The inclusive comparison handles the many ties induced by discrete graph statistics; the plus-one correction prevents invalid zero Monte Carlo p-values and supports finite-sample randomization validity (Phipson and Smyth 2010; Hemerik and Goeman 2018). The result reports the method, assignment count, ties, numerical tie tolerance, and attainable p-value resolution.

4.2. Multiplicity-controlled edge inference

For each $e \in \mathcal{E}$, an $m \times 2$ table contains present and absent counts. The default is a two-sided Fisher exact test. All $|\mathcal{E}|$ raw p-values form one prespecified family. Holm adjustment controls family-wise error under arbitrary dependence (Holm 1979). The BY option controls the false discovery rate under general dependence (Benjamini and Yekutieli 2001); BH is opt-in because its usual guarantee requires additional dependence conditions (Benjamini and Hochberg 1995).

For two groups, the signed effect is $p_{2,e} - p_{1,e}$ using constructor order. For more groups, the package reports $\max_k p_{k,e} - \min_k p_{k,e}$ and the groups attaining the extrema. The omnibus test does not imply pairwise significance.

4.3. Edge-separable computation and ablation

Equation 1 is a sum of additive edge contributions. For n graphs, $E = |\mathcal{E}|$ edges, and B assignments, the implementation stores an $n \times E$ graph-edge matrix and processes assignments in chunks. Group counts are matrix cross-products. The leading arithmetic cost is $O(mBnE)$, not $O(mBE)$; chunking limits temporary $E \times B$ allocations.

Edge separability also yields the package's main algorithmic optimization. A direct ranked-removal implementation recomputes a residual score for each of K candidate prefixes and each random assignment, with leading cost $O(KmBnE)$. **BrainNetTest** computes every assignment's edge contributions once and obtains all residual scores by cumulative subtraction. When $K = E$, the path-update stage falls from $O(BE^2)$ repeated tail sums to $O(BE)$ prefix sums. Section 8 checks numerical identity and measures both the chunked global engine and this prefix-sum update.

The optional ablation path sorts edges by raw p-value, decreasing absolute effect magnitude, and endpoints. The computational speedup is exact, but the statistical interpretation remains limited: because the order is selected from the observed labels, the returned `descriptive_tail_fraction` is not a post-selection p-value. No stopping set is returned, and non-rejection is not described as equivalence or statistical indistinguishability.

5. Software workflows

5.1. Two groups

The following complete example creates two synthetic groups and validates the input. The constructor is the single entry point that enforces the analysis contract, and its `print()` and `summary()` methods describe the data without echoing any adjacency matrix.

```
R> library("BrainNetTest")
R>
R> group_a <- generate_category_graphs(
+    n_graphs = 20, n_nodes = 10, n_communities = 2,
+    base_intra_prob = 0.80, base_inter_prob = 0.10, seed = 1
+ )
R> group_b <- generate_category_graphs(
+    n_graphs = 20, n_nodes = 10, n_communities = 2,
+    base_intra_prob = 0.45, base_inter_prob = 0.45, seed = 2
+ )
R>
R> networks <- brainnet_data(list(GroupA = group_a, GroupB = group_b))
R> networks

<brainnet_data>
2 groups; 10 nodes; 40 graphs
  GroupA: 20
  GroupB: 20

R> summary(networks)

Brain-network populations
  Nodes: 10
  Groups: 2
  GroupA: n = 20, mean density = 0.404
  GroupB: n = 20, mean density = 0.444
```

Passing the validated object to `brainnet_test()` runs both inferential layers and returns a classed `brainnet_result`.

```
R> result <- brainnet_test(
+    networks, n_permutations = 999, adjust = "holm", seed = 42
+ )
R> result

<brainnet_result>
Global permutation test: T = -29.32, p = 0.001 (Monte Carlo)
Edge inference: fisher + Holm; 1 of 45 edges selected

R> head(selected_edges(result)[, c(
+    "node1", "node2", "risk_difference", "p_adjusted"
+ )])
```

```
  node1 node2 risk_difference  p_adjusted
1     6     8           -0.65 0.001769117
```

The printed result does not dump stored matrices. `as.data.frame(result)` returns the complete edge table, `summary(result)` adds the highest-ranked edges, and `plot(result)` draws the classed result (Figure 1 uses the three-group result below).

5.2. Three groups

Multi-group input uses the same constructor and method dispatch. Edge p-values are omnibus; effects are observed ranges.

```
R> group_c <- generate_category_graphs(
+    n_graphs = 20, n_nodes = 10, n_communities = 2,
+    base_intra_prob = 0.25, base_inter_prob = 0.65, seed = 3
+  )
R> three_groups <- brainnet_data(
+    list(GroupA = group_a, GroupB = group_b, GroupC = group_c)
+  )
R> three_result <- brainnet_test(
+    three_groups, n_permutations = 999, adjust = "holm", seed = 43
+  )
R> three_result
```

```
<brainnet_result>
Global permutation test: T = -113.3, p = 0.001 (Monte Carlo)
Edge inference: fisher + Holm; 32 of 45 edges selected
```

The `plot()` method reuses the central graphs and selected edges stored in the result, so the original populations need not be supplied again (Figure 1).

```
R> plot(three_result, reference = "GroupC")
```

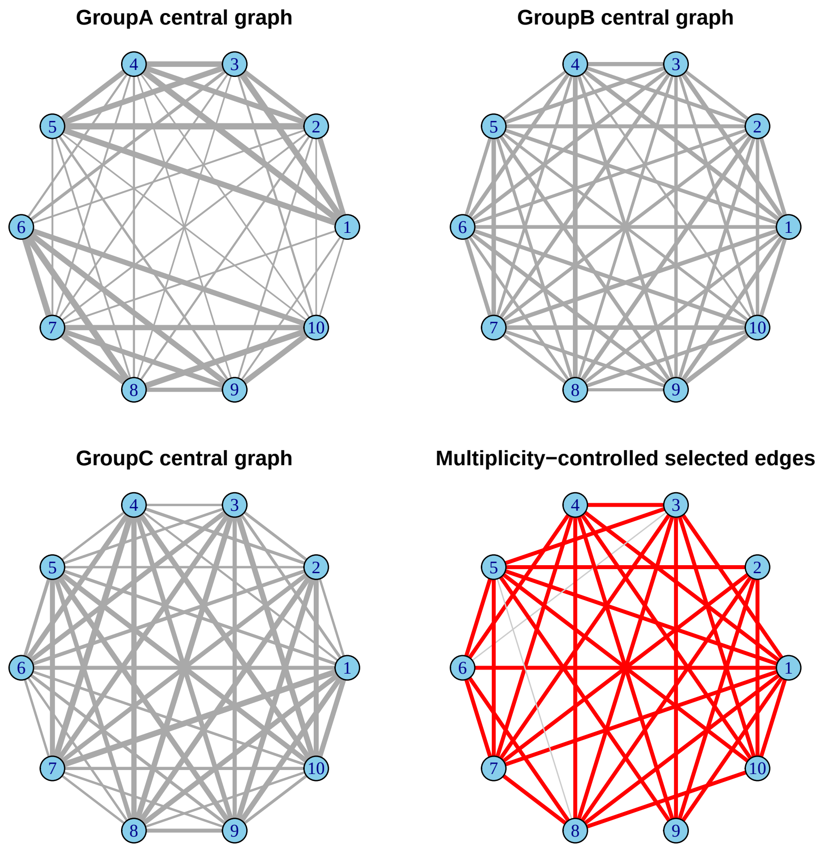

Figure 1: The `plot()` method applied to the three-group result. The first three panels show each group central graph, with edge width encoding the empirical edge frequency; the final panel highlights the multiplicity-controlled selected edges in red over the `GroupC` central graph.

6. Clinical application: Autism and resting-state connectivity

Autism spectrum disorder has been associated with altered resting-state functional connectivity, but findings can be distributed across many connections and are sensitive to site, motion, preprocessing, and thresholding (Di Martino, Yan, Li, Denio, Castellanos, Alaerts, Anderson, Assaf, Bookheimer, Dapretto *et al.* 2014). We use the openly shared, preprocessed resting-state fMRI derivatives of ABIDE I (Di Martino *et al.* 2014; Craddock, Benhajali, Chu, Chouinard, Evans, Jakab, Khundrakpam, Lewis, Li, Milham, Yan, and Bellec 2013) to ask a clinically motivated question: does diagnosis group correspond to a different spatial pattern of binary functional connections?

To avoid combining scanner sites, we use only the NYU sample and retain participants with mean framewise displacement below 0.2. For each participant we take mean time series from the AAL atlas (Tzourio-Mazoyer, Landeau, Papathanassiou, Crivello, Etard, Delcroix, Mazoyer, and Joliot 2002), form the Pearson correlation matrix, and keep the strongest 10%

of edges. Proportional thresholding gives every network the same edge count and avoids density confounding (van den Heuvel, de Lange, Zalesky, Seguin, Yeo, and Schmidt 2017); therefore the test concerns the anatomical distribution of edges, not total network density. The standalone `data-application.R` script downloads the public derivatives and records every construction choice.

```
R> connectomes <- readRDS("application/abide_connectomes.rds")
R> abide_full <- brainnet_data(
+    connectomes$populations,
+    node_labels = connectomes$node_labels
+  )
R> abide_balanced <- brainnet_data(
+    connectomes$balanced_populations,
+    node_labels = connectomes$node_labels
+  )
R> abide_full

<brainnet_data>
2 groups; 114 nodes; 171 graphs
  control: 98
  autism: 73

R> abide_balanced

<brainnet_data>
2 groups; 114 nodes; 110 graphs
  control: 55
  autism: 55

R> abide_full_result <- brainnet_test(
+    abide_full, n_permutations = 999, adjust = "holm", seed = 42
+  )
R> abide_balanced_result <- brainnet_test(
+    abide_balanced, n_permutations = 999, adjust = "holm", seed = 42
+  )
R> abide_full_result

<brainnet_result>
Global permutation test: T = -44.69, p = 0.001 (Monte Carlo)
Edge inference: fisher + Holm; 0 of 6441 edges selected

R> abide_balanced_result

<brainnet_result>
Global permutation test: T = -27.39, p = 0.01 (Monte Carlo)
Edge inference: fisher + Holm; 0 of 6441 edges selected
```

The full cohort contains 98 control and 73 autism participants. Sex composition and mean motion differ between these observational groups, so we also construct a deterministic 1:1 coarsened balance sample without using any network outcome: exact sex strata, five-year age bins, and 0.05 mean-displacement bins. This retains 55 participants per group (49 male and 6 female in each); mean ages are 14.25 and 14.75 years, and mean framewise displacements are 0.062 and 0.062.

The observed whole-graph statistic is extreme under label randomization in the full cohort ($T = -44.7$, $p = 0.001$) and in the balanced sensitivity sample ($T = -27.4$, $p = 0.01$). Thus, conditional on this site, preprocessing pipeline, atlas, and threshold, the data show a population-level association between diagnosis label and the spatial pattern of resting-state connections. Balancing the measured sex, age, and motion distributions does not remove that association. The balanced analysis of 110 networks, 114 regions, 6441 edge hypotheses, and 999 random assignments completes in under one second on the reference system.

Edge localization is more demanding. In the balanced sample, 3 edges have raw p-values below 0.001, but none survives Holm family-wise control or BY false-discovery control at $\alpha = 0.05$; the smallest Holm-adjusted p-value is 1.000. The strongest observed differences are reported for anatomical context, not as selected clinical biomarkers. Figure 2 displays their anatomical distribution and edge-wise evidence.

```
R> edges <- as.data.frame(abide_balanced_result)
R> edges$region1 <- connectomes$node_labels[edges$node1]
R> edges$region2 <- connectomes$node_labels[edges$node2]
R> top_edges <- head(edges[order(edges$p_value), c(
+    "region1", "region2", "risk_difference", "p_adjusted"
+  )], 5)
R> rownames(top_edges) <- NULL
R> top_edges
```

```
         region1        region2 risk_difference p_adjusted
1     Thalamus_L       Heschl_L       0.2545455          1
2   Precentral_L  Frontal_Sup_L      -0.3272727          1
3 Cingulum_Mid_R       Heschl_R      -0.2545455          1
4 Parietal_Sup_L Temporal_Mid_R      -0.1818182          1
5  Hippocampus_R Temporal_Mid_R       0.2727273          1
```

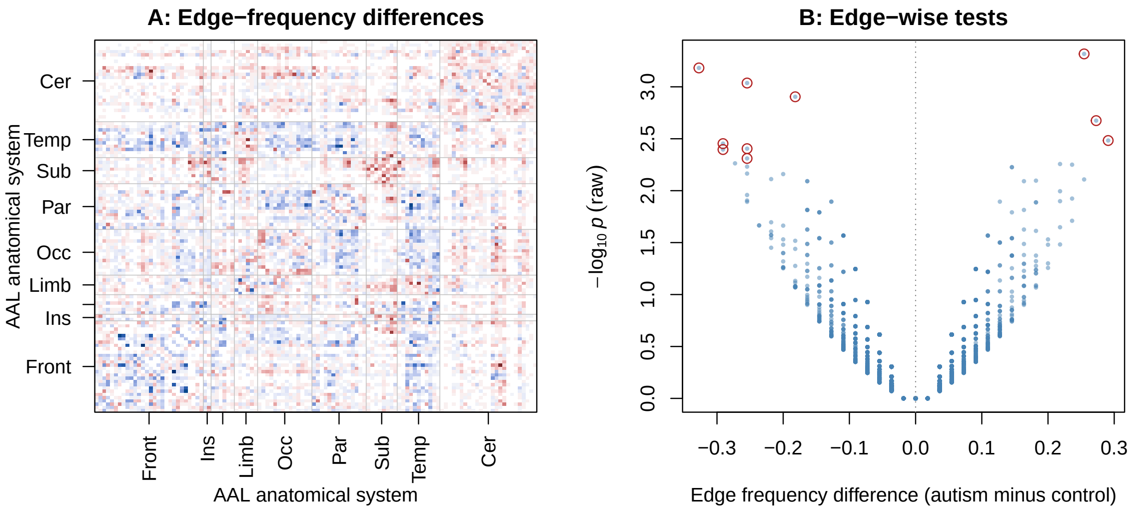

Figure 2: ABIDE clinical-group results in the balanced sensitivity sample. Panel A shows autism-minus-control differences in empirical edge frequency (blue: less frequent in autism; red: more frequent), with AAL regions grouped into broad anatomical systems. Panel B shows the same 6441 edges by effect and raw significance; the ten smallest raw p-values are circled. The global test detects a change in the distributed edge pattern, but no individual edge survives Holm or BY control.

This real-world result is clinically relevant but not diagnostic or causal. It is consistent with a distributed change in functional network organization for this NYU sample, while strict edge-level control prevents unsupported claims about single connections. Unmeasured confounding, the selected preprocessing pipeline, atlas, and threshold remain possible explanations. The application therefore shows both the value and the limits of the package: it reports a preprocessing-conditional population association and states clearly when localization is not supported.

7. Prespecified simulation validation

The inferential simulation cells and pass/fail criteria were frozen before the full run and are distributed in `simulation-design.md`; the separately disclosed prefix-sum timing benchmark was added during the brain-first manuscript restructuring and does not alter these cells. Every complete-null cell has 2000 replications; each partial-null cell has 1000. The complete-null gate required the one-sided 95% Wilson upper limit for both global rejection and Holm family-wise error to be at most 0.065. This is an implementation tolerance, not a proof of exactness.

7.1. Complete null

The six cells include independent edges, unequal sample sizes, positive within-graph dependence from latent density, negative dependence from fixed density, and sparse/dense three-group settings. Observations remain independent between subjects. Table 2 and Figure 3

scenario	global_rejection	global_upper_95	holm_fwer	holm_upper_95	by_fdr
dense_three_group	0.023	0.029	0.000	0.001	0.000
fixed_density	0.044	0.052	0.013	0.018	0.001
independent_balanced	0.046	0.054	0.010	0.014	0.002
independent_unbalanced	0.051	0.059	0.015	0.020	0.003
latent_density	0.040	0.048	0.006	0.010	0.001
sparse_three_group	0.030	0.037	0.000	0.001	0.000

Table 2: Complete-null calibration. Upper limits are one-sided 95% Wilson bounds.

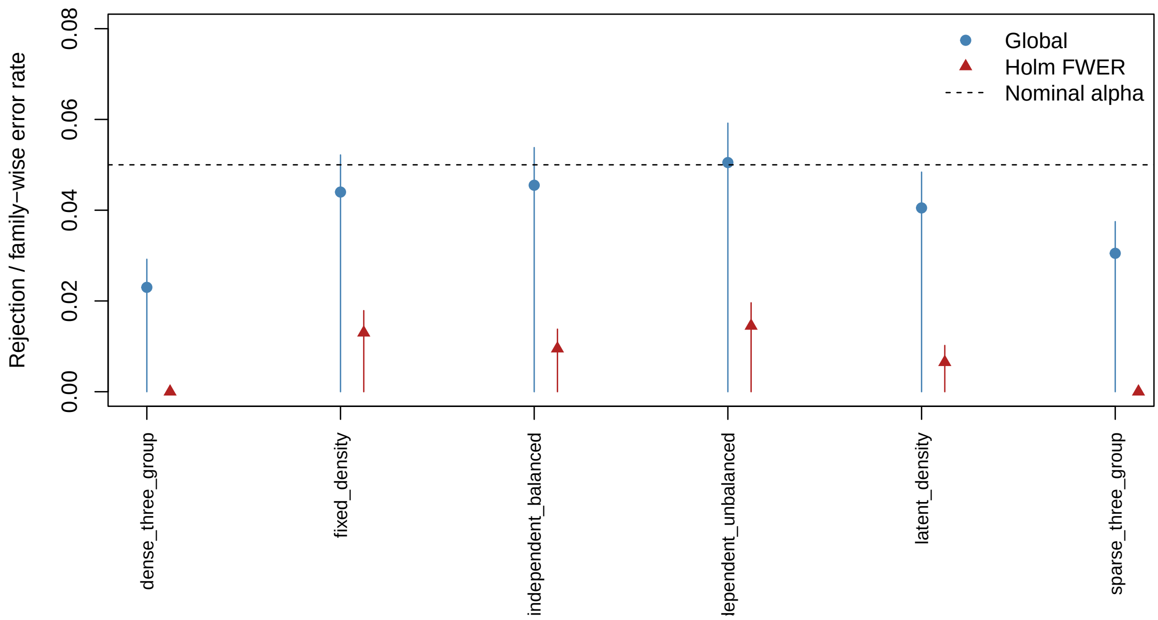

Figure 3: Global rejection and Holm family-wise error under the complete null. Vertical segments end at the prespecified one-sided 95% upper bounds; the dashed line is $\alpha = 0.05$.

summarize the calibration results.

7.2. Partial nulls and localization

Four alternatives cover localized positive shifts, scattered mixed-direction shifts, latent-density dependence among null edges, and three groups. Table 3 reports unconditional results, and Figure 4 compares localization rates; the separately labeled conditional table is included in the replication material. If R is the number selected, precision and false-discovery proportion divide by $\max(R, 1)$, so both are zero when no edge is selected.

Global power ranged from 0.953 to 0.998. Holm true-positive rate ranged from 0.171 to 0.202, while precision ranged from 0.676 to 0.750 and family-wise error from 0.005 to 0.011. The unadjusted comparator increased true-positive rate to 0.599–0.659 but raised family-wise error to 0.370–0.520. Thus multiplicity control is effective but materially conservative at 15–20 observations per group.

scenario	procedure	global_power	tpr	precision	fdp	fwer	jaccard	mean_selected
latent_dependent	BY	0.995	0.150	0.548	0.004	0.005	0.150	0.906
localized_positive	BY	0.994	0.141	0.518	0.002	0.005	0.141	0.853
scattered_mixed	BY	0.998	0.144	0.539	0.001	0.001	0.144	0.865
three_group	BY	0.953	0.106	0.414	0.003	0.003	0.106	0.638
latent_dependent	Holm	0.995	0.202	0.728	0.008	0.010	0.202	1.224
localized_positive	Holm	0.994	0.192	0.711	0.004	0.008	0.192	1.161
scattered_mixed	Holm	0.998	0.198	0.750	0.003	0.005	0.198	1.192
three_group	Holm	0.953	0.171	0.676	0.008	0.011	0.171	1.038

Table 3: Unconditional global power and edge-localization performance.

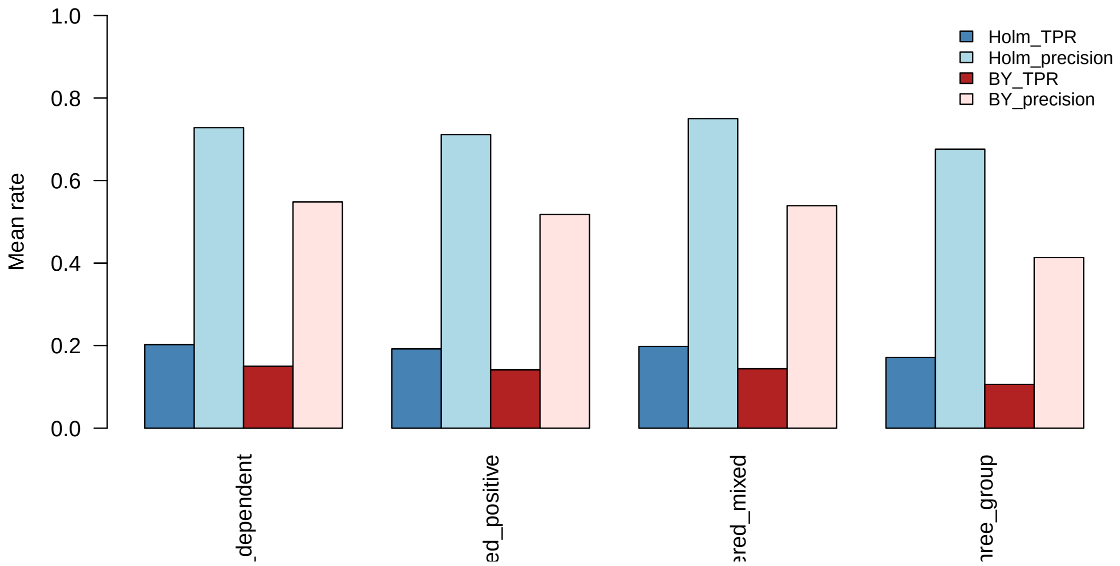

Figure 4: Mean true-positive rate and precision for Holm and BY selections across the four partial-null alternatives.

7.3. Scope stress test and sensitivity

The weak-null stress cell gives both groups marginal edge probability 0.5 but uses perfectly dependent edges in one group and independent edges in the other. Whole-graph exchangeability is false, so its global rejection rate is not a Type-I error estimate for the package's global null. Every edge-marginal null is true, however, so Holm family-wise error remains a required check. This cell makes the distinction between the two inferential layers explicit.

Observed global rejection was 0.065; this is descriptive because the whole-graph null is false. Holm family-wise error was 0.013, with one-sided 95% upper bound 0.018.

Sensitivity runs compare 99, 199, 999, and 4,999 random assignments on common data. Edge decisions are invariant to this setting because edge inference is separate from global randomization; global p-value resolution and elapsed time change as expected. Full cell-level results are in the replication tables.

Global power was 0.998 with 99 assignments and 1.000 with 4,999; the mean number of

Holm-selected edges remained 1.138 in every row. Median runtime rose from 0.012 to 0.026 seconds.

8. Computational performance

8.1. Chunked global randomization

The benchmark compares the chunked matrix engine with a direct loop over the same 499 assignment matrix. Equality of every assignment score is checked before timing is accepted. Each method is warmed up, measurement order is randomized, and fast calls are timed in calibrated batches. Table 4 reports medians over 10 repetitions; the largest measured median speedup was 19.1. Figure 5 shows scaling over the measured edge range.

n_nodes	n_edges	median_optimized_seconds	optimized_iqr	median_direct_seconds	median_speedup	input_megabytes
16.0000	120.0000	0.0009	0.0001	0.0174	19.0546	0.1207
32.0000	496.0000	0.0040	0.0004	0.0379	9.5315	0.2584
64.0000	2016.0000	0.0288	0.0029	0.1140	4.2772	0.8150
128.0000	8128.0000	0.1024	0.0117	0.4230	4.1230	3.0533
256.0000	32640.0000	0.4390	0.0498	1.7015	3.8508	12.0299

Table 4: Measured global-engine runtime and input-object size summaries.

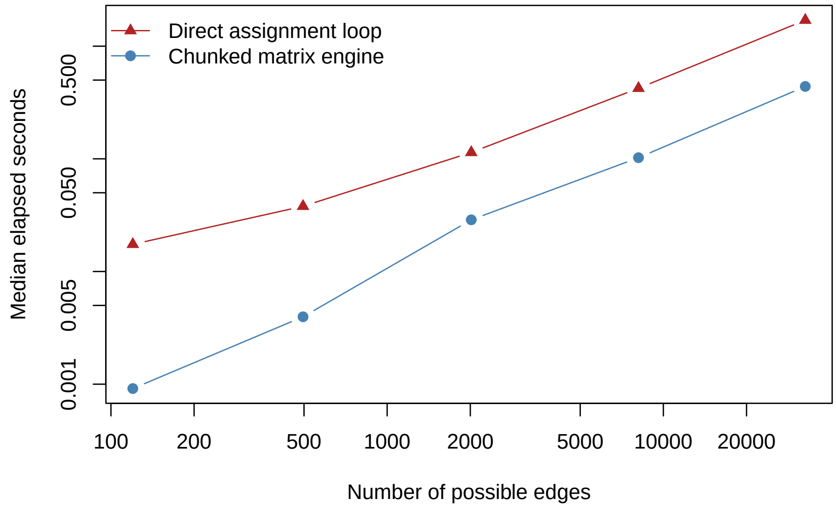

Figure 5: Measured runtime for direct and chunked assignment-score implementations. Both axes are logarithmic.

8.2. Prefix-sum ablation updates

The second benchmark isolates the ranked-removal update optimized by the prefix-sum algorithm. For each of 99 fixed assignment rows, both implementations receive the same edge-contribution vector and fixed edge order. The direct implementation repeatedly sums every remaining tail; the optimized implementation obtains all residual scores by one cumulative sum. Every complete path is checked for numerical equality before timing. Table 5 and Figure 6 report the update-stage results.

n_nodes	n_edges	median_prefix_seconds	prefix_iqr	median_repeated_tail_seconds	median_speedup
16.00000	120.00000	0.00013	0.00001	0.01031	80.50680
24.00000	276.00000	0.00023	0.00001	0.04244	180.05183
32.00000	496.00000	0.00035	0.00001	0.10067	283.31047
40.00000	780.00000	0.00052	0.00001	0.23500	449.70640
48.00000	1128.00000	0.00073	0.00005	0.43100	591.05946
56.00000	1540.00000	0.00096	0.00002	0.73500	755.48817
64.00000	2016.00000	0.00125	0.00012	1.18500	952.61768

Table 5: Measured runtime for repeated-tail and prefix-sum ablation updates. The contribution matrix is computed before both timings.

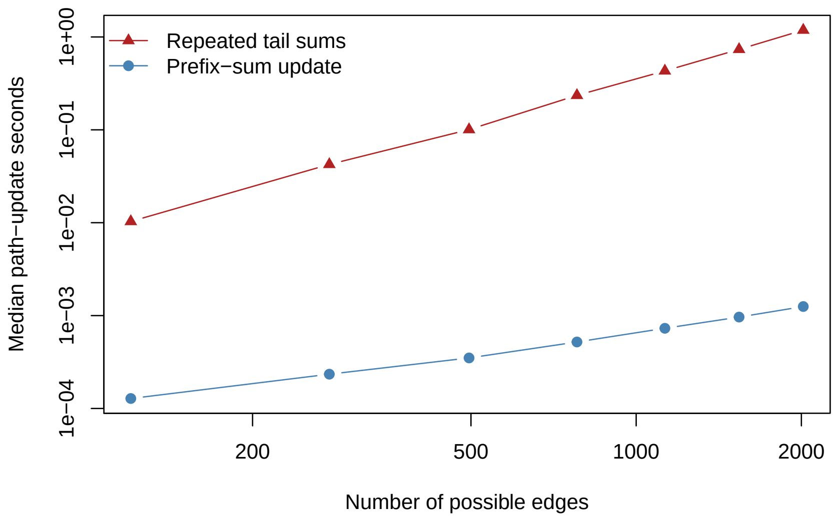

Figure 6: Measured runtime of the ablation path-update stage. Both methods produce identical residual paths; both axes are logarithmic.

Across the measured range, the largest median update-stage speedup was 953. This is deliberately not reported as an end-to-end speedup: edge ranking and the assignment-by-edge contribution calculation are common to both methods. Together, the two benchmarks show that the software engineering contribution covers both the main randomization engine and the iterative diagnostic inherited from the original package.

The benchmarks do not extrapolate unobserved multi-day runtimes. Performance depends on hardware, BLAS, graph density, and the chosen edge test; full session information and raw timings are distributed.

9. Discussion

BrainNetTest 1.0.0 turns the Fraiman and Fraiman brain-network test into a validated clinical connectomics workflow. The classed interface makes malformed connectome populations fail early, standard methods make results inspectable without list plumbing, and the result records the randomization and multiplicity information needed for interpretation. Global inference, edge-level inference, and adaptive ablation now have distinct names and contracts.

The ABIDE application demonstrates real-world use rather than only a simulated example. In both the full cohort and the sex-, age-, and motion-balanced sensitivity sample, the observed global statistic is extreme under autism-control label randomization. No individual edge survives strict multiplicity control. Clinically, this is a useful distinction: the data support a preprocessing-conditional population association, but they do not support a single-edge biomarker claim. The result is not causal or diagnostic, and it remains tied to the selected sample, preprocessing pipeline, atlas, and threshold.

The computational contribution is also retained. Chunked matrix products accelerate randomization scores, while exact prefix-sum updates replace repeated edge-removal calculations. These are software-engineering improvements to a statistical workflow, not substitutes for inferential validation; Section 8 therefore reports both numerical identity and measured speed.

The supported domain remains precise: aligned binary undirected brain networks from independent participants. The global score is sensitive to marginal edge-frequency patterns rather than every possible distributional change. Holm control can be conservative when thousands of connections are tested, while BY trades additional conservatism for false-discovery control under general dependence. The same mathematical contract can be useful outside neuroimaging, but brain-network population analysis is the package's primary application.

Future work should add restricted randomization for family, site, paired, and longitudinal designs; covariate-adjusted edge models; principled support for weighted or directed networks; and external validation across clinical cohorts. Such extensions require new contracts and validation rather than coercion inside the current API.

Computational details

The package requires R 4.0.0 or later and imports **stats**, **graphics**, and **igraph** (Csardi and Nepusz 2006). Unit tests use **testthat** (Wickham 2011); the vignettes and this manuscript are built with **knitr** (Xie 2026) and **rmarkdown** (Allaire, Xie, Dervieux, McPherson, Luraschi, Ushey, Atkins, Wickham, Cheng, Chang, and Iannone 2026). The authoritative `code-full.R` script regenerates every reported simulation and benchmark result, table, figure, and numeric macro. The `data-application.R` script downloads the openly shared ABIDE derivatives used in Section 6 and saves the derived binary connectomes that the manuscript reads; it is the only component that needs network access. A separate standalone `code.R` and generated `code.html` exercise a reduced pipeline within the reviewer-time guideline. Full `sessionInfo()` output is supplied in `generated/sessionInfo.txt`.

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